HW8

Greyson Knippel
MATH 339 — Gonzaga University
compiled with LaTeX

November 8, 2026

4.1 - 17

$$T\left(\begin{bmatrix}x\\y\end{bmatrix}\right)=\begin{bmatrix}-x\\y\end{bmatrix},\qquad \mathbf{u}=\begin{bmatrix}-2\\3\end{bmatrix},\qquad \mathbf{v}=\begin{bmatrix}2\\-2\end{bmatrix}.$$

a.

T negates the first component and keeps the second, so:

$$T(\mathbf{u})=\begin{bmatrix}2\\3\end{bmatrix},\qquad T(\mathbf{v})=\begin{bmatrix}-2\\-2\end{bmatrix}.$$

b.

$$\mathbf{u}+\mathbf{v}=\begin{bmatrix}0\\1\end{bmatrix}\quad\Rightarrow\quad T(\mathbf{u}+\mathbf{v})=\begin{bmatrix}0\\1\end{bmatrix},$$

$$T(\mathbf{u})+T(\mathbf{v})=\begin{bmatrix}2\\3\end{bmatrix}+\begin{bmatrix}-2\\-2\end{bmatrix}=\begin{bmatrix}0\\1\end{bmatrix}=T(\mathbf{u}+\mathbf{v}).$$

so yes, $T(\mathbf{u}+\mathbf{v})=T(\mathbf{u})+T(\mathbf{v})$

c.

one example working isn't enough, so we check both properties for vectors $\begin{bmatrix}x_1\\y_1\end{bmatrix},\begin{bmatrix}x_2\\y_2\end{bmatrix}$ and scalar c:

$$T\left(\begin{bmatrix}x_1+x_2\\y_1+y_2\end{bmatrix}\right)=\begin{bmatrix}-x_1-x_2\\y_1+y_2\end{bmatrix}=\begin{bmatrix}-x_1\\y_1\end{bmatrix}+\begin{bmatrix}-x_2\\y_2\end{bmatrix}=T\left(\begin{bmatrix}x_1\\y_1\end{bmatrix}\right)+T\left(\begin{bmatrix}x_2\\y_2\end{bmatrix}\right),$$

$$T\left(c\begin{bmatrix}x\\y\end{bmatrix}\right)=\begin{bmatrix}-cx\\cy\end{bmatrix}=c\begin{bmatrix}-x\\y\end{bmatrix}=cT\left(\begin{bmatrix}x\\y\end{bmatrix}\right),$$

so yes, T is a linear transformation because its a reflection across the y axis.

4.1 - 27

$$T(x^2)=2x-1,\qquad T(-3x)=x^2-1.$$

since T is linear, we can pull the scalar out of $T(-3x)$:

$$T(-3x)=-3\,T(x)=x^2-1\quad\Rightarrow\quad T(x)=-\tfrac{1}{3}(x^2-1)=-\tfrac{1}{3}x^2+\tfrac{1}{3}.$$

but we still know nothing about $T(1)$

a.

using linearity,

$$T(2x^2 - 3x + 2) = 2\,T(x^2) - 3\,T(x) + 2\,T(1).$$

this needs $T(1)$, and 1 is not a linear combination of x^2 and x, so the given information doesn't determine it. so no, $T(2x^2 - 3x + 2)$ cannot be determined

b.

$3x^2 - 4x$ only uses x^2 and x, so we can write it in terms of the given inputs:

$$3x^2 - 4x = 3(x^2) + \tfrac{4}{3}(-3x).$$

then by linearity,

$$T(3x^2 - 4x) = 3\,T(x^2) + \tfrac{4}{3}\,T(-3x) = 3(2x - 1) + \tfrac{4}{3}(x^2 - 1) = 6x - 3 + \tfrac{4}{3}x^2 - \tfrac{4}{3} = \tfrac{4}{3}x^2 + 6x - \tfrac{13}{3}.$$

so yes, $T(3x^2 - 4x) = \frac{4}{3}x^2 + 6x - \frac{13}{3}$.